

are a special type of NP problem defined by the feature that every single NP problem can be quickly reduced to an NP-complete problem; in other words, if you can solve one NP-complete problem, you can solve every NP problem. If you can solve any single NP-complete problem in polynomial time, you can solve every NP problem in polynomial time, which would prove that $P = NP$. As it happens, sudoku and the n-queens completion problem are both NP-complete. This means that finding a polynomial-time algorithmic solution to either of them would not only solve every existing NP problem but also earn you a million dollars and worldwide, lifelong fame (not to mention the power to beat everyone you know in friendly sudoku competitions).

A proof that $P \neq NP$ would probably not be as straightforward as a solution to sudoku. The notion that $P \neq NP$ means that there are NP problems that cannot be solved by any algorithm with polynomial runtime. Proving this amounts to proving a negative, and it is conceptually much harder to prove that something cannot exist than it is to point to an example of something. Making progress in a proof that $P \neq NP$ will require extended study in theoretical computer science beyond the scope of this book. Though this path is harder, it seems to be the consensus among researchers that $P \neq NP$, and that if there is ever a resolution to the P versus NP question, it will probably be a proof that $P \neq NP$.

The P versus NP question is not the only deep mystery related to algorithms, although it is the most immediately lucrative one. Every aspect of the field of algorithm design has wide-open fields for adventurers to charge into. There are not only theoretical and academic questions, but also practical ones related to how to implement algorithmically sound practices in business contexts. Waste no time: remember what you have learned here and sally forth anon, carrying your new skills with you to the utmost bounds of knowledge and practice, on your lifelong algorithmic adventure. Friends, adieu.